

Offer blockchain-based traceability and visibility solutions for capital-constrained recyclers in closed-loop supply chains.

Wholesale × Product traceability and digital product passport × Blockchain and DLT · OS098 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
41 is the world moving	13 can we play, can we win	LOW 1 named in the evidence	€2.1bn modelled evidence · per year	LATER research and standards activi...	L3 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Wholesale, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity offers blockchain-based traceability and visibility solutions to help capital-constrained recyclers secure financing by providing transparent, verifiable data on their operations. It targets SMEs in the recycling sector that struggle to obtain credit due to a lack of visibility, and it exists now because blockchain technology has matured to enable practical, interoperable traceability solutions.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€3.5bn €706m – €15.1bn	€2.1bn €420m – €9.0bn	€2.6m €525k – €11.2m	modelled

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

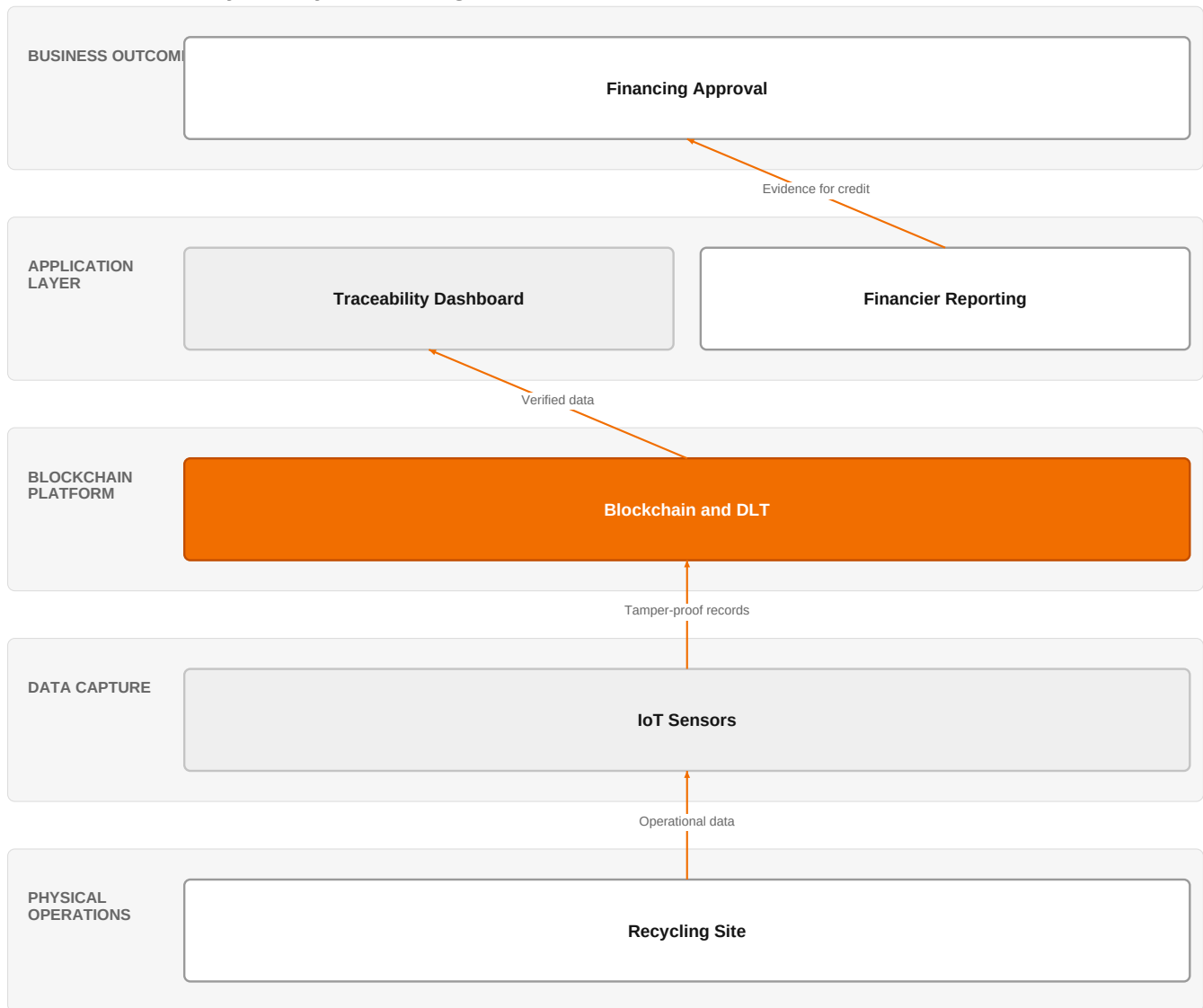
Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Wholesale (EU27_2020)	148,208 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Blockchain and DLT	60.4 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	18,053 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (CN); the estimate covers the European footprint only.

The solution, in one picture

Blockchain Traceability for Recycler Financing



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This solution provides financiers with tamper-evident proof of recycling operations, enabling capital-constrained recyclers to secure financing.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Capital-constrained recyclers, especially SMEs, have long faced financial problems and struggle to secure financing for recycling initiatives due to credit issues. Blockchain-enabled traceability and visibility can address this by providing the transparency needed to build trust with financiers. Recent research highlights blockchain's potential to enhance transparency, traceability, and trust in supply chains, addressing challenges such as fraud, inefficiency, and lack of visibility. Additionally, the integration of blockchain with IoT and big data is enabling real-time, tamper-proof data sharing across supply chain networks, making such solutions more feasible and effective.

Sources: SIG-C5698801304D [openalex.org 2026-04-03](https://openalex.org/2026-04-03); SIG-EC14886FDE80 [openalex.org 2026-03-22](https://openalex.org/2026-03-22); SIG-A85B5F017CD1 [openalex.org 2026-02-27](https://openalex.org/2026-02-27); SIG-25647612A208 [openalex.org 2025-08-30](https://openalex.org/2025-08-30)

Who buys, and why now

The buyer is likely the CFO or head of sustainability at a mid-sized recycling company, while the operational pain is felt by the operations director who struggles to demonstrate the provenance and quality of recycled materials to potential financiers. The trigger is a rejected loan application or a financing round that falls through due to lack of verifiable data on recycling volumes and processes. They need to show lenders that their operations are legitimate, efficient, and low-risk.

Why it is hot now — every claim bound to a dated source

- Blockchain-enabled traceability helps capital-constrained recyclers secure financing by providing visibility into their operations. [SIG-C5698801304D]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts to deploy sensors and devices at recycling sites to capture operational data, and its Blockchain and DLT technology to create an immutable, transparent ledger of recycling activities. Orange Group researchers would contribute to designing the blockchain architecture and ensuring interoperability. The engagement would start with a pilot project to demonstrate the value to financiers, then scale to a full deployment with SOC 2 Type II certification to assure data security and compliance.

Why Orange can win

Orange can win because it combines IoT expertise with blockchain technology, enabling end-to-end traceability from physical operations to digital records. Its SOC 2 Type II certification provides a trust signal for handling sensitive data. However, the assets are thin: there is no specific recycling domain expertise or pre-built solution, so Orange would need to partner or develop domain knowledge to compete effectively.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

LOW competitive intensity (score 1.4; 1 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitors include hyperscalers like AWS, which offer blockchain services and IoT platforms, and global system integrators like Accenture, Capgemini, and NTT DATA, which have deep supply chain consulting expertise. Telecom peers such as Deutsche Telekom, Telia, and Vodafone Business also offer IoT and connectivity solutions. Actility, as an industrial IoT specialist, could provide LoRaWAN-based tracking. The customer will compare Orange's ability to integrate IoT and blockchain with the scale and ecosystem of hyperscalers and the consulting depth of integrators.

Sources: SIG-01499FFBAF54 arxiv.org 2026-06-05

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	evidenced	SIG-01499FFBAF54
Accenture	Global systems integrator	competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	competes in OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	competes in OX: Smart Industries	structural	—
Telia Company	Telecom operator peer	competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	competes in OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently demonstrate the provenance and quality of your recycled materials to potential financiers?
- 2 What specific data points would lenders need to see to approve a loan for your recycling operations?
- 3 Have you ever been denied financing because of a lack of verifiable operational data?
- 4 What is your current level of digitalization in tracking materials through your recycling process?
- 5 Are you under pressure from customers or regulators to provide more transparency in your supply chain?
- 6 Would you be open to a pilot project that uses IoT sensors and blockchain to create a verifiable record of your recycling activities?

Objections you will meet

They will say	You can answer
We don't have the budget for blockchain and IoT solutions.	We understand that capital is tight. That's why we propose a pilot that focuses on the most critical data points, and we can help you articulate the ROI to financiers who may be willing to fund the implementation as part of a loan package.
Our operations are too small for such a complex solution.	That's exactly the point. Small and medium recyclers are often overlooked because they lack visibility. Our solution is designed to be scalable, starting with a minimal viable product that can grow with your business.
Will lenders actually trust blockchain data?	It's a fair concern. Blockchain is still new, but it provides tamper-evident records that can increase trust. We can work with you to engage lenders early and demonstrate how the data meets their requirements.

Risks and unknowns

The main risk is that the recycling SMEs may lack the digital maturity to adopt blockchain-based solutions, and the financing community may not yet recognize blockchain-verified data as a credit enhancement. Additionally, the regulatory landscape for digital product passports is still evolving, and the cost of IoT deployment may be prohibitive for capital-constrained SMEs. It is unknown whether the value proposition will resonate with financiers in practice.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that positions Orange's Blockchain and DLT technology, backed by IoT experts and SOC 2 Type II certification, as the foundation for a digital product passport tailored to capital-constrained recyclers.
Sales	In a conversation with a wholesale distributor's sustainability lead, say: 'We have SOC 2 Type II certified infrastructure and IoT expertise that could underpin a traceability solution for your closed-loop supply chain partners.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's IoT experts and Group researchers can jointly prototype a blockchain-based digital product passport for recyclers, focusing on the financing-visibility angle.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-C5698801304D	2026-04-03	openalex.org	1	Blockchain-enabled traceability and visibility: Financing for a capital-constrained recycler in...
SIG-6B094A3F3A46	2026-03-26	openalex.org	1	Sybil attack defense in blockchain-based industrial IoT systems using decentralized federated l...
SIG-EC14886FDE80	2026-03-22	openalex.org	1	A Blockchain-Enabled Smart Contract Architecture for Enhancing Transparency, Traceability, and ...
SIG-85F36B96E175	2026-03-01	openalex.org	1	The Integration of RFID and Blockchain Technology for the Supply Chain Traceability of Durians
SIG-A85B5F017CD1	2026-02-27	openalex.org	1	Blockchain and Big Data Integration for Transparent and Traceable Supply Chains
SIG-1C2398920027	2026-02-04	openalex.org	1	Enhancing fruit supply chain traceability through blockchain and cryptographic protocols for ac...
SIG-D9CA278082C5	2025-11-27	openalex.org	1	A Blockchain-Based Approach to Enhancing Traceability and Transparency in the Supply Chain
SIG-14060280DE3E	2025-11-06	openalex.org	1	Enhancing Agricultural Supply Chain Traceability with Blockchain, Smart Contracts, and E-Labell...
SIG-98BCC6F19BE4	2025-10-27	openalex.org	1	QuantumTrust-FedChain: A Blockchain-Aware Quantum-Tuned Federated Learning System for Cyber-Res...
SIG-25647612A208	2025-08-30	openalex.org	1	A Narrative Review and Gap Analysis of Blockchain for Transparency, Traceability, and Trust in ...
SIG-4674EE3A0873	2025-02-26	The Role of Network Secur...	1	5G-Enabled Smart Healthcare System with the Integration of Blockchain Technology
SIG-01499FFBAF54	2026-06-05	arxiv.org	3	Blockchain Infrastructure for Intelligent Cyber--Physical--Social Systems:Post-Quantum Security...
SIG-4B7C038778C3	2026-05-18	arxiv.org	3	Operationalising Post Quantum TLS Automated Configuration Profiling and Hybrid PQC Deployment i...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:57+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.